PUBPOL 205 · FALL 2026

Problem Set 5 Solutions

Chapter 17 (externalities) and Chapters 6–7 (production and costs). Due in class Oct 13, 2026.

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CHAPTER 17 · EXTERNALITIES

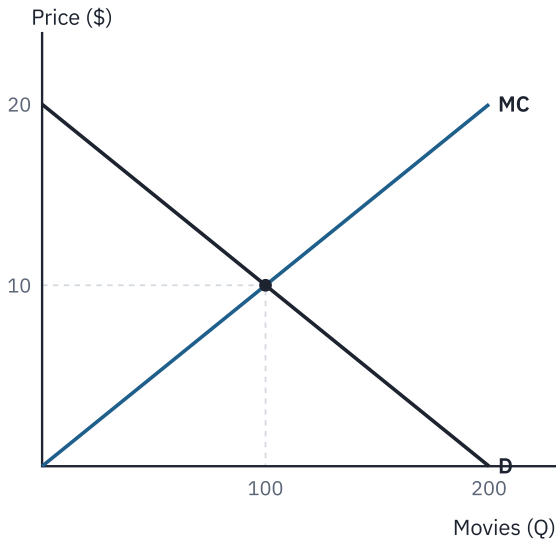
Problem 1 17.8 Drive-in movies

Private demand $P = 20 - 0.1Q$; marginal cost $MC = 0.1Q$; external marginal benefit $EMB = 2 - 0.01Q$.

a. Set private demand equal to marginal cost:

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20 − 0.1Q = 0.1Q  ⇒  Q = 100
P = 20 − 0.1(100) = $10
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The market shows 100 movies at $10.

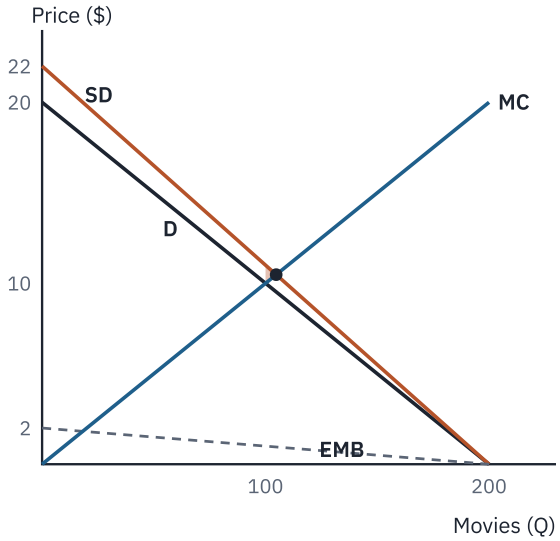


Private equilibrium: Q = 100, P = $10.

b. The social demand curve adds the external marginal benefit vertically to private demand:

```
SD = D + EMB = (20 − 0.1Q) + (2 − 0.01Q) = 22 − 0.11Q
```

EMB runs from \$2 at Q = 0 to \$0 at Q = 200. SD runs from \$22 at Q = 0 to \$0 at Q = 200.



Social demand SD = D + EMB. The efficient point (dot) is at Q ≈ 105. The deadweight loss is the small shaded sliver between SD and MC from Q = 100 to Q ≈ 105.

c. The Movie Czar sets social demand equal to marginal cost:

```
22 − 0.11Q = 0.1Q   ⇒   Q = 22/0.21 ≈ 105
P = 22 − 0.11(105) ≈ $10.45
```

Optimal quantity ≈ 105 movies at a price of about \$10.45.

d. The deadweight loss is the triangle between SD and MC from the private output (100) to the efficient output (≈105), shaded in the graph in (b). The answer key computes:

```
DWL = ½ × (105 − 100) × (10.45 − 9.50) = $2.375
```

> Using the hint directly: at Q = 100 the last unit's external benefit is EMB = 2 − 0.01(100) = \$1, so the triangle's height is SD − MC = 11 − 10 = \$1 and its base is 104.76 − 100 = 4.76. DWL = ½ × 4.76 × 1 ≈ \$2.38. Same answer to rounding.

e. Yes. Privately run theaters underproduce because they ignore the benefit to viewers outside the fence. Government can raise output to the socially optimal level. It could also reach the same result with a price-based policy, such as a per-movie subsidy to private producers equal to the EMB at the optimum.

Problem 2 17.9 Leather

Inverse demand $P = 50 - 0.5Q$; private MC $= 0.45Q$; external marginal cost $EMC = 0.05Q$.

a. Social marginal cost is $SMC = MC + EMC = 0.45Q + 0.05Q = 0.5Q$. The efficient output sets MB = SMC:

$$50 - 0.5Q = 0.5Q \;\Rightarrow\; Q^* = 50$$
$$P = 50 - 0.5(50) = \$25$$

Set the quota at 50 units. The market price with the quota is $25.

b. The tax equals the external damage at the efficient output: $EMC(50) = 0.05 \times 50 =$ **$2.50 per unit**. With the tax, producers' marginal cost becomes $MC = 0.45Q + 2.50$. Producers set MB = MC:

$$50 - 0.5Q = 0.45Q + 2.50 \;\Rightarrow\; Q = 50$$
$$P = 50 - 0.5(50) = \$25$$

50 units are bought and sold. Buyers pay $25; sellers keep $25 − $2.50 = $22.50.

> For comparison, the unregulated market sets $50 - 0.5Q = 0.45Q$, giving $Q \approx 52.6$, so the policy trims output by about 2.6 units.

Problem 3 17.11 Genetic testing

a. A positive externality. The person who buys the test gets private benefits: ancestry, links to relatives. Law enforcement also gets potential links to suspects, and the more people who are tested, the more matches there are. That benefit goes to law enforcement (and society), not to the buyer.

b. Markets underprovide goods that carry positive externalities. Buyers act on their private marginal benefit, which lies below the social marginal benefit, so the market quantity falls short of the efficient quantity. There is likely too little genetic testing.

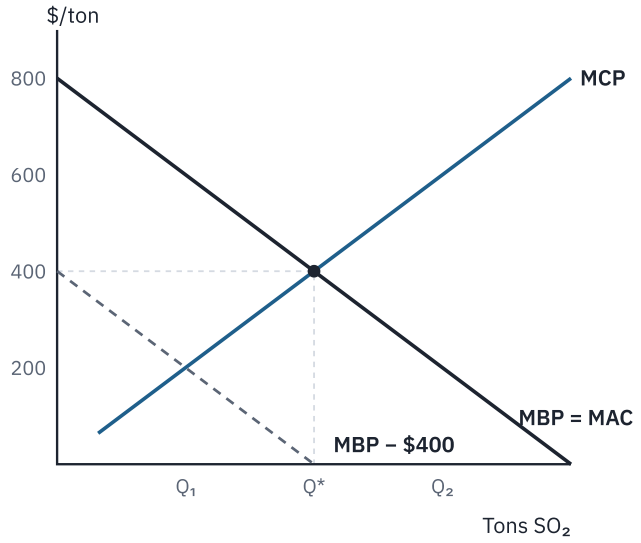
c. The goal is more testing, which rules out taxes and cap-and-trade; both reduce quantity. A quota (requiring testing) would raise output but raises serious constitutional and privacy concerns. That leaves a **subsidy**, which raises the incentive to be tested without compelling anyone.

Problem 4 17.13 Sulfur dioxide

From the diagram: MBP = MAC falls from \$800 to \$0; MCP (external marginal cost of pollution) rises; they cross at \$400.

a. When pollution is already low, cutting one more ton is expensive: the cheap abatement options are used up. When pollution is high, the first tons can be cut cheaply. So the marginal abatement cost, read as the marginal benefit to the firm of emitting one more ton, falls as emissions rise.

b. The efficient level of pollution is where MBP = MAC crosses MCP, at a marginal value of \$400 per ton.



Efficient pollution where MBP = MCP = \$400. With a \$400 tax, firms emit until MBP – 400 = 0, i.e. at Q*. Q_1 and Q_2 mark outcomes under \$600 and \$200 taxes.

c. Firms ignore MCP, so they would emit until MBP = 0. A \$400 per-ton tax shifts their net marginal benefit of polluting down by \$400 (the dashed line in the figure). Firms now emit until MBP = \$400, which happens exactly at the efficient level, because the MCP there is \$400.

d. **\$600 is too high:** firms emit until MBP = \$600, which happens at a quantity below the efficient level (Q_1). Society gives up abatement-cost savings worth more than the damage avoided.

\$200 is too low: firms emit until MBP = \$200, a quantity above the efficient level (Q_2). Those extra tons cause damage greater than their value to firms.

Problem 5 6.6 Hawg Wild

K \ L	1	2	3	4	5	6
1	100	132	155	174	190	205
2	152	200	235	264	289	310
3	193	255	300	337	368	396
4	230	303	357	400	437	470
5	263	347	408	457	500	538
6	293	387	455	510	558	600

a. Long-run. Both labor and capital vary in the table. In a short-run production function capital is held fixed.

b. **347 diners (K = 5, L = 2).**

c.

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MP of nephew = Q(K=5, L=3) − Q(K=5, L=2) = 408 − 347 = 61
```

d. No. Diminishing marginal product of labor is defined holding capital fixed. Along any row, the MP of labor falls as L rises. Comparing MP across rows changes capital, and more capital makes each worker more productive; that is not a violation.

e.

```
MP of smoker = Q(K=6, L=2) − Q(K=5, L=2) = 387 − 347 = 40
```

f. Compare marginal product per dollar:

```
MPL / w = 61 / 12 = 5.083
MPK / r = 40 / 8  = 5.000
```

Hire the worker: labor yields slightly more output per dollar at the margin.

Problem 6 6.9 Fetzer valves

At the point where the isoquants cross, both countries use the same labor–capital combination. Now remove a fixed amount of capital. Along the steeper isoquant (Country A), output is maintained with a smaller increase in labor than along the flatter one (Country B). Labor substitutes for capital more effectively in A, so A's MRTS = MPL/MPK is higher, consistent with more productive, more skilled labor.

Country A is the United States; Country B is China.

Problem 7 6.13 Isocost lines

Wage w = \$12 per hour; rental rate r = \$7 per hour.

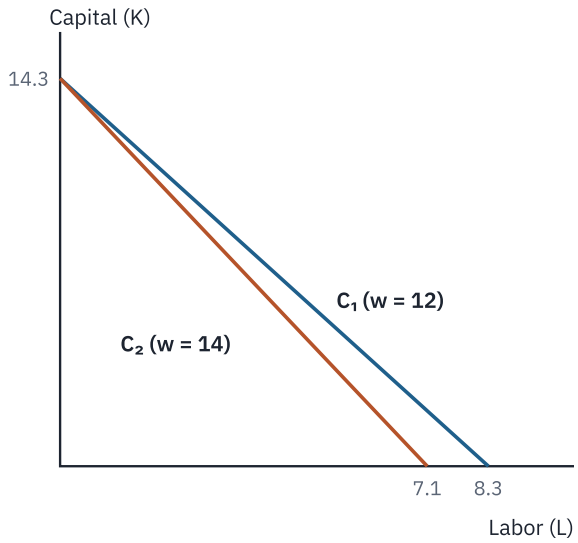
a.

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TC = 12L + 7K
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b.

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100 = 12L + 7K  ⇒  K = 100/7 − (12/7)L
```

c.



C_1: slope −12/7. C_2 after the wage rises to \$14: slope −2.

d. Vertical intercept K = 100/7 ≈ 14.3: the capital \$100 rents if no labor is hired. Horizontal intercept L = 100/12 ≈ 8.3: the labor \$100 hires if no capital is rented.

e. Slope = −w/r = −12/7 ≈ −1.71. It is the market rate at which capital trades for labor: one more worker costs the same as 12/7 units of capital.

f.

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100 = 14L + 7K  ⇒  K = 100/7 − 2L
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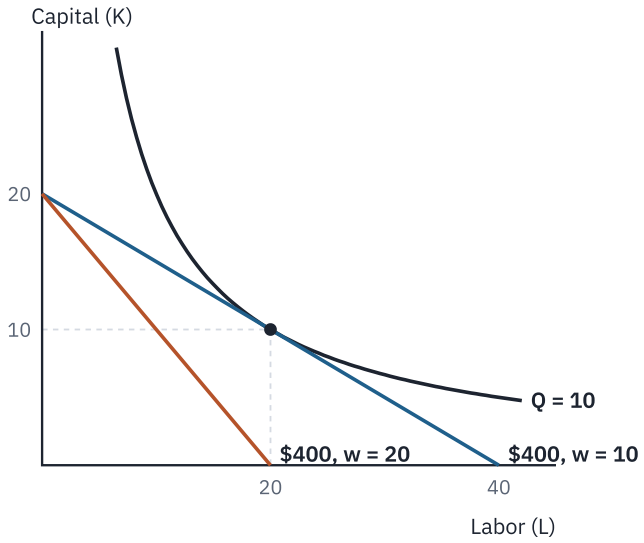
The isocost line rotates inward around the vertical intercept (100/7 is unchanged) and becomes steeper, with slope −2. The horizontal intercept falls to 100/14 ≈ 7.1. Line C_2 in the graph above.

Problem 8 6.17 Creekwood Lawn and Landscape

Cost-minimizing at L = 20, K = 10, Q = 10 lawns, C = \$400. The \$400 isocost hits L = 40 and K = 20.

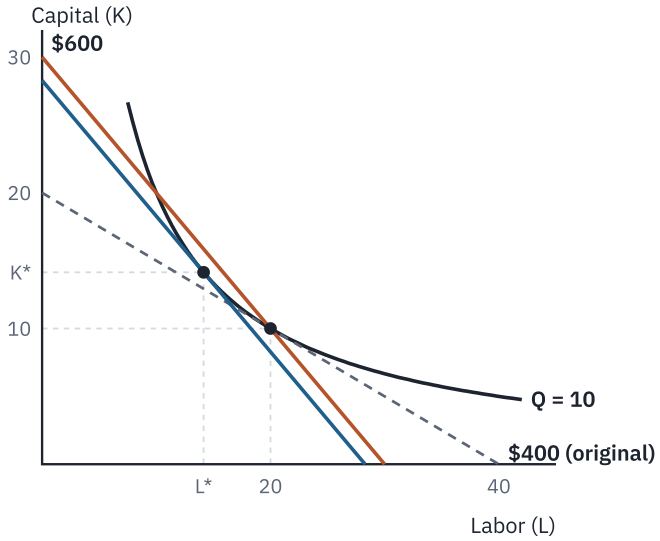
a. \$400 buys at most 40 workers (horizontal intercept), so w = 400/40 = **\$10/hour**. It buys at most 20 units of capital (vertical intercept), so r = 400/20 = **\$20/hour**.

b. At w = \$20, \$400 buys at most 20 workers. The \$400 isocost rotates inward around K = 20 to the L-intercept at 20. It no longer reaches the Q = 10 isoquant, so serving 10 lawns for \$400 is impossible. **Creekwood's cost of serving its customers rises.**



After the wage doubles, the \$400 isocost no longer reaches Q = 10.

c. The original bundle now costs 20 × \$20 + 10 × \$20 = \$600. The \$200 subsidy covers the difference, so Creekwood pays \$400 out of pocket for exactly its old inputs. The new isocost (total spending \$600) passes through (L = 20, K = 10) with the steeper slope −w/r = −1, so it hits K = 30 and L = 30.



Orange: the \$600 isocost after the wage increase, through the original bundle (20, 10); the subsidy covers \$200 of it. Blue: the cheaper isocost tangent to Q = 10 at (L*, K*), with less labor and more capital. Dashed: the original \$400 isocost. Isoquant shape is illustrative.

d. No. Labor is now relatively more expensive, and the subsidy arrives whether or not Creekwood spends it on labor. The steeper isocost through (20, 10) cuts the isoquant rather than being tangent to it, so Creekwood can lower costs by substituting toward capital: it moves to the point L* < 20, K* > 10 where an isocost with slope −1 is tangent to Q = 10.

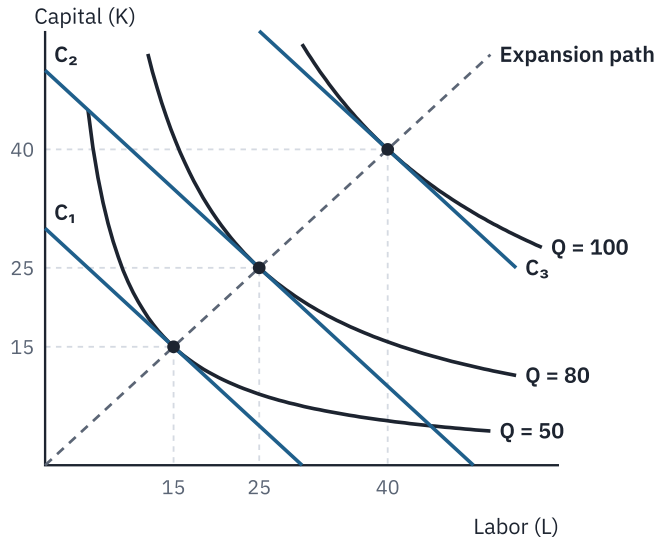
Problem 9 6.23 Cam's Cannolis

w = r = \$10, so every isocost line has slope −w/r = −1.

a. The isocost lines tangent to each isoquant give the cost-minimizing bundles:

Q = 50: L = 15, K = 15 · Q = 80: L = 25, K = 25 · Q = 100: L = 40, K = 40.

b. The expansion path connects these tangency points: (15, 15), (25, 25), (40, 40). Here it is the 45° ray from the origin.

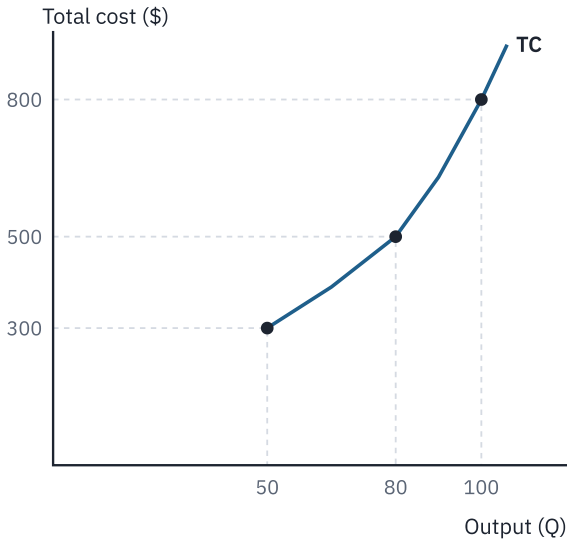


Isocosts with slope −1 tangent to each isoquant. Isoquant shapes are illustrative.

c.

Output	K	L	r	w	TC = rK + wL
50	15	15	10	10	300
80	25	25	10	10	500
100	40	40	10	10	800

d.



Total cost at the three cost-minimizing points.

Total cost rises faster than output (from 80 to 100 units, 20 more units cost $300), so the curve is convex over this range.

Problem 10 7.14 Funnel cake costs OPTIONAL

Identities used: TC = FC + VC; MC = ΔTC; AVC = VC/Q; AFC = FC/Q; ATC = TC/Q. Gray cells were legible in the original.

Steps: AFC(5) = 11 gives FC = 55 at every Q. VC(1) = 17 and MC(2) = 15 give VC(2) = 32. TC(3) = 101 gives VC(3) = 46. AVC(4) = 14.5 gives VC(4) = 58. TC(5) = 122 matches VC(5) = 67. ATC(6) = 21 gives TC(6) = 126.

Q	TC	FC	VC	MC	AVC	AFC	ATC
0	55	55	0	—	—	—	—
1	72	55	17	17	17	55	72
2	87	55	32	15	16	27.5	43.5
3	101	55	46	14	15.33	18.33	33.67
4	113	55	58	12	14.5	13.75	28.25
5	122	55	67	9	13.4	11	24.4
6	126	55	71	4	11.83	9.17	21

Solutions follow the course answer key. Graphs redrawn to scale from the given equations.